Introduction

The Mann-Whitney U test (equivalent to the Wilcoxon rank-sum test) compares two independent groups using ranks rather than raw values. It is the non-parametric analogue of the two-sample t-test, and is appropriate for ordinal outcomes or continuous data that clearly violate normality.

Prerequisites

Ranks, two-sample comparison.

Theory

For independent samples $X_1, \dots, X_{n_1}$ and $Y_1, \dots, Y_{n_2}$, pool the observations, rank them, and sum the ranks in the first group to get R_1. The U statistic is

$$U_1 = R_1 - \frac{n_1(n_1+1)}{2}.$$

Under $H_0 : F_X = F_Y$, U_1 has a known distribution computed via the Wilcoxon rank sum. Large-sample normal approximation uses

$$E[U] = n_1 n_2/2, \quad \mathrm{Var}(U) = n_1 n_2(n_1+n_2+1)/12.$$

Effect sizes:

- **Rank-biserial correlation** $r_{rb} = 1 - 2U/(n_1 n_2)$, range $[-1, 1]$.
- **Probability of superiority** $A = U/(n_1 n_2)$.

Assumptions

- Two independent groups.
- Observations at least ordinal.
- Similar distribution shapes for a median-difference interpretation; otherwise the test compares distributions more generally.

R Implementation

```r
library(effectsize)
set.seed(2026)

group1 <- rlnorm(25, meanlog = log(50), sdlog = 0.4)
group2 <- rlnorm(30, meanlog = log(60), sdlog = 0.4)

wilcox.test(group1, group2)

# Effect size: rank-biserial
rank_biserial(group1, group2)

# Median IQR comparisons
c(median1 = median(group1), IQR1 = IQR(group1),
  median2 = median(group2), IQR2 = IQR(group2))
```

Output & Results

```
        Wilcoxon rank sum test with continuity correction
W = 240, p-value = 0.008

r (rank biserial) | 95% CI
----------------------------
-0.36             | [-0.59, -0.10]
```

Significant difference at p = 0.008; rank-biserial -0.36 indicates a moderate shift with group 1 tending to be smaller.

Interpretation

"Serum biomarker concentrations were significantly lower in the control group than in cases (Mann-Whitney $U = 240$, $p = 0.008$, rank-biserial $r = -0.36$). Median concentration was 47 (IQR 34-59) ng/mL vs. 62 (IQR 47-73) in cases."

Practical Tips

- Report medians and IQRs alongside the test; means and SDs are inappropriate when you chose Mann-Whitney.
- For small samples with no ties, use the exact distribution (`exact = TRUE`); with ties, R issues a warning and falls back to the normal approximation.
- The test is insensitive to outliers but sensitive to distributional shape differences (not just medians).
- For paired ordinal data, use the Wilcoxon signed-rank test.
- Power relative to the t-test is about 95 % under normality and can exceed it under heavy tails.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests